

Question 1 | Correct | Mark 1.00 out of 1.00

Given an array A of sorted integers and another non negative integer k, find if there exists 2 indices i and j such that $A[j] - A[i] = k$, $i \neq j$.

Input Format:

First Line n - Number of elements in an array

Next n Lines - N elements in the array

k - Non - Negative Integer

Output Format:

1 - If pair exists

0 - If no pair exists

Explanation for the given Sample Testcase:

YES as $5 - 1 = 4$

So Return 1.

For example:

Input	Result
3 1 3 5 4	1

Answer: (penalty regime: 0 %)

```

1  #include <stdio.h>
2
3  int main() {
4      int n, k;
5      scanf("%d", &n);
6
7      int a[n];
8
9      for (int i = 0; i < n; i++)
10         scanf("%d", &a[i]);
11
12     scanf("%d", &k);
13
14     int i = 0, j = 1;
15
16     while (j < n) {
17         int diff = a[j] - a[i];
18
19         if (diff == k) {
20             printf("1");
21             return 0;
22         }
23         else if (diff < k) {
24             j++;
25         }
26         else {
27             i++;
28         }
29
30         if (i == j)
31             j++;
32     }
33
34     printf("0");
35
36     return 0;
37 }
```

	Input	Expected	Got	
✓	3 1 3 5 4	1	1	✓
✓	10 1 4 6 8 12 14 15 20 21 25 1	1	1	✓
✓	10 1 2 3 5 11 14 16 24 28 29 0	0	0	✓
✓	10 0 2 3 7 13 14 15 20 24 25 10	1	1	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

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